

Relationship Between Mole Fraction and Mass

MultXeu Gradients

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Derivation

Let x_i be the mole fraction of component i , M_i its molar mass, and m_i its mass. The mole fraction is defined as

$$x_i = \frac{n_i}{\sum_j n_j}, \quad (1)$$

where the number of moles of component i is

$$n_i = \frac{m_i}{M_i}. \quad (2)$$

Substituting this into the expression for mole fraction gives

$$x_i = \frac{\frac{m_i}{M_i}}{\sum_j \frac{m_j}{M_j}}. \quad (3)$$

This is the direct relationship between the mass of each component and its mole fraction. Alternatively, if we start from mass fractions $w_i = \frac{m_i}{m_{\text{tot}}}$, then

$$x_i = \frac{\frac{w_i}{M_i}}{\sum_j \frac{w_j}{M_j}}. \quad (4)$$

Using $w_i = \frac{m_i}{m_{\text{tot}}}$ and $w_j = \frac{m_j}{m_{\text{tot}}}$, we obtain

$$x_i = \frac{\frac{m_i/m_{\text{tot}}}{M_i}}{\sum_j \frac{m_j/m_{\text{tot}}}{M_j}}. \quad (5)$$

Factoring out the common term $1/m_{\text{tot}}$ from numerator and denominator gives

$$x_i = \frac{\frac{1}{m_{\text{tot}}} \left(\frac{m_i}{M_i} \right)}{\frac{1}{m_{\text{tot}}} \sum_j \frac{m_j}{M_j}}. \quad (6)$$

The total mass cancels, leaving the same result:

$$x_i = \frac{\frac{m_i}{M_i}}{\sum_j \frac{m_j}{M_j}}. \quad (7)$$

Solving for Mass

Rearranging the equation to isolate m_i gives

$$m_i = x_i M_i \sum_j \frac{m_j}{M_j}. \quad (8)$$

This relation is useful when the mole fraction and molar masses are known, and one wants to recover the corresponding mass contribution of component i .